

CASE STUDY

Additional Information for: DLBDSME01 – Model Engineering

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1. DEVELOPMENT OF AN INTERPRETABLE MODEL FOR BREAST CANCER PREDICTION

1.1 Filename

- folder: data for case 1
- data.csv: file with a list of ids, labels and features to classify breast cancer

1.2 Columns / attribute information

The following numbers correspond to the respective columns in the file.

- 1) ID number
- 2) Diagnosis (Labels: M = malignant, B = benign)
- 3-32) Ten real-valued features are computed for each cell nucleus:
 - a) radius (mean of distances from center to points on the perimeter)
 - b) texture (standard deviation of gray-scale values)
 - c) perimeter
 - d) area
 - e) smoothness (local variation in radius lengths)
 - f) compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
 - g) concavity (severity of concave portions of the contour)
 - h) concave points (number of concave portions of the contour)
 - i) symmetry
 - j) fractal dimension ("coastline approximation" - 1)

The mean (extension „_mean“), standard error (extension „_se“) and "worst" or largest mean (mean of the three largest values, extension „_worst“) of these features were computed for each image, resulting in 30 features.

All feature values are recoded with four significant digits.

Missing attribute values: none

Class distribution: 357 benign, 212 malignant

1.3 Further information about the dataset

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image. The 3-dimensional space is described in K.P. Bennett & O.L. Mangasarian, 1992. The data can also be found on UCI Machine Learning Repository:

<https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+%28Diagnostic%29>

Introductory literature:

K. P. Bennett & O. L. Mangasarian (1992). Robust Linear Programming Discrimination of Two Linearly Inseparable Sets. *Optimization Methods and Software 1*, 23-34.

2. DEVELOPMENT OF A DEMAND FORECASTING MODEL FOR PUBLIC TRANSPORT

2.1 Filename

- folder: Data for task 2
- data-apr14.csv, data-may14.csv, data-jun14.csv, data-jul14.csv, data-aug14.csv, data-sep14.csv: files containing a time stamp, GPS data and a base station code.
- data-on-map.py: a python script to visualize the data on a New York City map and save the map as “map.html”, which can be viewed in any web browser.

Note: You do not need to use this file. It is only an example that shows how data could be visualized on a map. Other visualizations are also valid. If used, to run the script you need to have Python (<https://www.python.org/>), pandas (<https://pandas.pydata.org/>) and Folium (<https://python-visualization.github.io/folium/>) installed. Moreover, the script assumes to be executed in the same folder where the data is.

2.2 Columns / attribute information

The following numbers correspond to the respective column numbers in each file.

- 1) Date / Time: The date and time of the vehicle pickup
- 2) Lat: The latitude of the pickup
- 3) Lon: The longitude of the pickup
- 4) Base: The code of the base associated with the respective vehicle.

2.3 Further information

The data was extracted from the following dataset:

<https://www.kaggle.com/datasets/fivethirtyeight/uber-pickups-in-new-york-city>

3. DEVELOPMENT OF A SPAM FILTER

3.1 Filename

- folder: Data for task 3
- SMSSpamCollection.csv: collection of SMS labeled as spam or no spam.

3.2 Columns / attribute information

The following numbers correspond to the respective column numbers in the file.

- 1) Column1: Label of the corresponding short message (spam, ham = no spam).
- 2) Column2: Text of the corresponding short message.

3.3 Further information

This dataset can be obtained from <http://archive.ics.uci.edu/ml/datasets/sms+spam+collection>

The following information is a compilation from the readme file in the download folder:

The SMS Spam Collection v.1 (hereafter the corpus) is a set of SMS tagged messages that have been collected for SMS Spam research. It contains one set of SMS messages in English of 5,574 messages, tagged according being ham (legitimate) or spam.

Compilation:

This corpus has been collected from free or free for research sources at the Web:

- A collection of between 425 SMS spam messages extracted manually from the Grumbletext Web site. This is a UK forum in which cell phone users make public claims about SMS spam messages, most of them without reporting the very spam message received. The identification of the text of spam messages in the claims is a very hard and time-consuming task, and it involved carefully scanning hundreds of web pages. The Grumbletext Web site is: <http://www.grumbletext.co.uk/>
- A list of 450 SMS ham messages collected from Caroline Tag's PhD Theses available at <http://etheses.bham.ac.uk/253/1/Tagg09PhD.pdf>
- A subset of 3,375 SMS ham messages of the NUS SMS Corpus (NSC), which is a corpus of about 10,000 legitimate messages collected for research at the Department of Computer Science at the National University of Singapore. The messages largely originate from Singaporeans and mostly from students attending the University. These messages were collected from volunteers who were made aware that their contributions were going to be made publicly available. The NUS SMS Corpus is available at: <http://www.comp.nus.edu.sg/~rpnlpir/downloads/corpora/smsCorpus/>
- The amount of 1,002 SMS ham messages and 322 spam messages extracted from the SMS Spam Corpus v.0.1 Big created by José María Gómez Hidalgo and public available at: <http://www.esp.uem.es/jmgomez/smsspamcorpus/>

Introductory literature:

Almeida, T.A., Gómez Hidalgo, J.M., Yamakami, A. (2011). *Contributions to the study of SMS Spam Filtering: New Collection and Results*. Proceedings of the 2011 ACM Symposium on Document Engineering (ACM DOCENG'11), Mountain View, CA, USA.